

entrega

March 10, 2025

```
[42]: import pandas as pd
import numpy as np
simulaciones = pd.read_json("simulation.jsonlines", lines=True)
planes = pd.read_json("plans.jsonlines", lines=True)
```

1 Paso 2: Transformación de los datos

1. Aplico la función proporcionada en la página de Github

De este modo puedo ver con más claridad como están ordenados los datos, elimino la columna truck porque ya tenemos los datos en diferentes columnas.

```
[43]: planes = planes.join(planes.trucks.explode().apply(pd.Series), lsuffix='_sim').
↪reset_index(drop=True).drop("trucks", axis=1)
planes
```

```
[43]:                                     items_sim \
0      [{'itemId': '0', 'locationId': '326ae8dc7810de...
1      [{'itemId': '0', 'locationId': '326ae8dc7810de...
2      [{'itemId': '0', 'locationId': '326ae8dc7810de...
3      [{'itemId': '0', 'locationId': '326ae8dc7810de...
4      [{'itemId': '0', 'locationId': '326ae8dc7810de...
..
635    [{'itemId': '0', 'locationId': 'd1214b27823e6d...
636    [{'itemId': '0', 'locationId': 'd1214b27823e6d...
637    [{'itemId': '0', 'locationId': 'd1214b27823e6d...
638    [{'itemId': '0', 'locationId': 'd1214b27823e6d...
639    [{'itemId': '0', 'locationId': 'd1214b27823e6d...

                                     simulationId \
0      a2bfd344-3b14-462a-82e7-d42aca54a650
1      a2bfd344-3b14-462a-82e7-d42aca54a650
2      a2bfd344-3b14-462a-82e7-d42aca54a650
3      a2bfd344-3b14-462a-82e7-d42aca54a650
4      a2bfd344-3b14-462a-82e7-d42aca54a650
..
635    6deec7b7-6c11-418c-a1bd-f99a9e3261b6
636    6deec7b7-6c11-418c-a1bd-f99a9e3261b6
```

```

637 6deec7b7-6c11-418c-a1bd-f99a9e3261b6
638 6deec7b7-6c11-418c-a1bd-f99a9e3261b6
639 6deec7b7-6c11-418c-a1bd-f99a9e3261b6

```

```

                                items \
0    [23, 24, 4, 25, 26, 27, 28, 40, 41, 42, 43, 13...
1    [74, 75, 76, 67, 68, 69, 70, 92, 93, 94, 55, 5...
2    [18, 19, 57, 58, 88, 89, 90, 91, 54, 20, 21, 2...
3                                [87, 0, 1, 2, 3, 33, 95, 96, 97]
4                                [31, 32, 37, 38, 39, 66]
..                                ...
635  [53, 54, 55, 83, 84, 85, 86, 26, 27, 29, 30, 3...
636  [72, 73, 74, 88, 89, 90, 91, 43, 62, 63, 64, 3...
637                                [6, 7, 8, 9]
638  [44, 45, 46, 47, 94, 20, 21, 22, 95, 96, 97, 6...
639  [79, 80, 81, 82, 24, 25, 40, 41, 42, 12, 23, 9...

```

```

                                route truck_id
0    [{'destination': 'af3ea5c0e98151a51bd39c64', '...' 5534TPH
1    [{'destination': 'faf06210580833f8c7963506', '...' 2423VRT
2    [{'destination': 'bef2df0471ee1c26b4ab6ba9', '...' 0265TYL
3    [{'destination': '7a249403ad22a72495769f75', '...' 8809GPH
4    [{'destination': '1e38734ff239ccf75bf04b95', '...' 1161GJN
..                                ...
635  [{'destination': '225cd814768e8ca390983313', '...' 5534TPH
636  [{'destination': '417bf3aae83127cd905f3bc9', '...' 3953RLD
637  [{'destination': '1e1251e186dd507f63e4d1e6', '...' 5030LXK
638  [{'destination': '6a33d0bcd3624ccb86febf09', '...' 3321FBL
639  [{'destination': '17c23e21622baec6f08861dd', '...' 6270NFM

```

[640 rows x 5 columns]

```
[44]: planes = planes.groupby(["simulationId", "truck_id"]).agg(list)
```

2. Ahora busco sacar información del dataframe de simulaciones

Observo que hay 8 tipos de eventos diferentes, “Truck departed” indica desde donde sale el camión al comienzo, “Truck initialized” indica cuando inicia el trabajo, “Truck received packets” indica la lista de paquetes indicando los paquetes y el lugar al que hay que enviarlos, “Truck arrived” indica a qué ubicación ha llegado, “Truck started delivering” indica el lugar al que está repartiendo y el paquete en concreto, “Truck ended delivering” indica cuándo ha terminado de repartir un paquete en concreto, “Truck departed to depot” indica el final de la jornada y “Truck ended route” indica que ha llegado al final de la ruta.

```
[45]: simulaciones = simulaciones.rename(columns={"truckId": "truck_id"})
```

```
[46]: lista_eventos = ["Truck departed", "Truck arrived", "Truck started_
↳delivering", "Truck ended delivering"]
```

```
[47]: simulaciones = simulaciones[simulaciones.eventType.isin(lista_eventos)]
```

```
[48]: simulaciones = simulaciones.groupby(["simulationId", "truck_id"]).agg(list)
```

3. Junto los dos dataframes

```
[49]: conjunto = planes.join(simulaciones, lsuffix="_plan", rsuffix="_simul")
```

```
[50]: conjunto
```

```
[50]:      items_sim \
simulationId      truck_id
007f3d08-14a7-4a93-af9e-f0d9cfbcea94 3321FBL  [[{'itemId': '0', 'locationId':
'bc451e2cea161...
                                     3953RLD  [[{'itemId': '0', 'locationId':
'bc451e2cea161...
                                     5030LXK  [[{'itemId': '0', 'locationId':
'bc451e2cea161...
                                     5534TPH  [[{'itemId': '0', 'locationId':
'bc451e2cea161...
                                     6270NFM  [[{'itemId': '0', 'locationId':
'bc451e2cea161...
...
...
fea13535-ade6-4215-96ca-dab5b4ef309b 0013DYS  [[{'itemId': '0', 'locationId':
'17d843cfd0ff5...
                                     3953RLD  [[{'itemId': '0', 'locationId':
'17d843cfd0ff5...
                                     5534TPH  [[{'itemId': '0', 'locationId':
'17d843cfd0ff5...
                                     6270NFM  [[{'itemId': '0', 'locationId':
'17d843cfd0ff5...
                                     6965XLY  [[{'itemId': '0', 'locationId':
'17d843cfd0ff5...

      items \
simulationId      truck_id
007f3d08-14a7-4a93-af9e-f0d9cfbcea94 3321FBL  [[78, 79, 50, 51, 52, 23, 21, 58,
59, 60, 61, ...
                                     3953RLD
[[44, 45]]
                                     5030LXK      [[16, 17, 18, 19, 80, 81,
48, 38, 39, 22]]
                                     5534TPH  [[68, 69, 70, 53, 54, 55, 40, 41,
42, 43, 73, ...
                                     6270NFM      [[6, 7, 8, 35, 36, 37, 62,
63, 64, 65, 57]]
...
```

```

...
fea13535-ade6-4215-96ca-dab5b4ef309b 0013DYS
[[4, 5]]

3953RLD [[71, 72, 73, 74, 53, 38, 39, 18,
70, 66, 67, ...

5534TPH [[61, 62, 63, 11, 12, 13, 14, 60,
19, 20, 21, ...

6270NFM [[36, 37, 9, 10, 17, 16, 56, 25,
26, 27, 28, 2...

6965XLY [[15, 44, 45, 46, 47, 22, 23, 24,
2, 3, 40, 41...

```

```

route \
simulationId truck_id
007f3d08-14a7-4a93-af9e-f0d9cfbcea94 3321FBL [[{'destination':
'2e8d1b7b9fad4f083892d83c', ...

3953RLD [[{'destination':
'4bc556d929cc052029e14a78', ...

5030LXK [[{'destination':
'0550fd3ee7831440c5cf21c5', ...

5534TPH [[{'destination':
'3a471470f7f1d508af6cdfdb', ...

6270NFM [[{'destination':
'04463c1410164bb7855a830f', ...
...
...
fea13535-ade6-4215-96ca-dab5b4ef309b 0013DYS [[{'destination':
'a840e3e2d217fe732e00f08c', ...

3953RLD [[{'destination':
'9492562782bb6f6befb7dfc8', ...

5534TPH [[{'destination':
'5a555ff3e81103468a36f757', ...

6270NFM [[{'destination':
'4a600d1ab9d6a40ea60dd088', ...

6965XLY [[{'destination':
'58bb022022a0e97ef0efcf46', ...

```

```

eventDescription \
simulationId truck_id
007f3d08-14a7-4a93-af9e-f0d9cfbcea94 3321FBL [(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...

3953RLD [(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...

5030LXK [(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...

5534TPH [(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...

```

Madrid, [-3.704186...	6270NFM	[(plaza del Dos de Mayo, 6,
...		
...		
fea13535-ade6-4215-96ca-dab5b4ef309b	0013DYS	[(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...		
	3953RLD	[(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...		
	5534TPH	[(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...		
	6270NFM	[(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...		
	6965XLY	[(plaza del Dos de Mayo, 6,
Madrid, [-3.704186...		
eventTime \		
simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	[660, 751620, 751620, 872250,
872250, 992820, ...		
	3953RLD	[1500, 4637550, 4637550, 4758120,
4758150, 481...		
	5030LXK	[570, 231690, 231720, 352290,
352290, 472920, ...		
	5534TPH	[1380, 793020, 793020, 913620,
913620, 974220,...		
	6270NFM	[630, 776820, 776820, 837420,
837450, 958050, ...		
...		
...		
fea13535-ade6-4215-96ca-dab5b4ef309b	0013DYS	[1050, 3902280, 3902280, 3962880,
3962880, 402...		
	3953RLD	[540, 253380, 253380, 313980,
313980, 374580, ...		
	5534TPH	[690, 724680, 724680, 845280,
845280, 905910, ...		
	6270NFM	[720, 577680, 577680, 638280,
638280, 758880, ...		
	6965XLY	[690, 421380, 421380, 481980,
482250, 2697810,...		
eventType		
simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	[Truck departed, Truck started
delivering, Tru...		
	3953RLD	[Truck departed, Truck started
delivering, Tru...		
	5030LXK	[Truck departed, Truck arrived,

```

Truck started ...
                                5534TPH  [Truck departed, Truck arrived,
Truck started ...
                                6270NFM  [Truck departed, Truck arrived,
Truck started ...
...
...
fea13535-ade6-4215-96ca-dab5b4ef309b 0013DYS [Truck departed, Truck arrived,
Truck started ...
                                3953RLD  [Truck departed, Truck arrived,
Truck started ...
                                5534TPH  [Truck departed, Truck arrived,
Truck started ...
                                6270NFM  [Truck departed, Truck arrived,
Truck started ...
                                6965XLY  [Truck departed, Truck arrived,
Truck started ...

[640 rows x 6 columns]

```

4. Vuelvo a aplicar la misma función pero esta vez a la columna route, obteniendo la duración del trayecto desde un punto a otro, también elimino la columna route

```
[51]: conjunto.route.explode().apply(pd.Series).head()
```

```

[51]:      0  \
simulationId      truck_id
007f3d08-14a7-4a93-af9e-f0d9cfbcea94 3321FBL  {'destination':
'2e8d1b7b9fad4f083892d83c', 'd...
                                3953RLD  {'destination':
'4bc556d929cc052029e14a78', 'd...
                                5030LXK  {'destination':
'0550fd3ee7831440c5cf21c5', 'd...
                                5534TPH  {'destination':
'3a471470f7f1d508af6cdfdb', 'd...
                                6270NFM  {'destination':
'04463c1410164bb7855a830f', 'd...

      1  \
simulationId      truck_id
007f3d08-14a7-4a93-af9e-f0d9cfbcea94 3321FBL  {'destination':
'f5a3aab0c10fb7c5c60f2b01', 'd...
                                3953RLD  {'destination':
'ce6feed06bc1a9e5aff95f04', 'd...
                                5030LXK  {'destination':
'87c216f2b62e7b2f643ad7d4', 'd...
                                5534TPH  {'destination':
'331afadf3cd2b8fa97eeca2', 'd...

```

	6270NFM	{'destination':
'7fbe65343adc2eb786272bfa', 'd...		

2 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'4b5f692a806a39d1317d15d4', 'd...		

3953RLD

NaN

5030LXK	{'destination':
---------	-----------------

'74fe1f5b828bddbbbd194a8f', 'd...

5534TPH	{'destination':
---------	-----------------

'05915a11e99fca75d7480069', 'd...

6270NFM	{'destination':
---------	-----------------

'c529324b91395971fb21755c', 'd...

3 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'64bba266de127a5949f52d9a', 'd...		

3953RLD

NaN

5030LXK	{'destination':
---------	-----------------

'66a73c577ad48cc9719b7381', 'd...

5534TPH	{'destination':
---------	-----------------

'0983b6c023b45c47425f456d', 'd...

6270NFM	{'destination':
---------	-----------------

'8fc6257ad9efa4296a7d3a17', 'd...

4 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'6942f2624e971761b7dd9208', 'd...		

3953RLD

NaN

5030LXK	{'destination':
---------	-----------------

'527ee974e61483070ff0ea1e', 'd...

5534TPH	{'destination':
---------	-----------------

'0bf195334776828444a19f08', 'd...

6270NFM	{'destination':
---------	-----------------

'ce6feed06bc1a9e5aff95f04', 'd...

5 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'103daf6394133e32f804e175', 'd...		

3953RLD

NaN	5030LXK	{'destination':
'ce6feed06bc1a9e5aff95f04', 'd...		
	5534TPH	{'destination':
'3d1a2178e4ccce5da9ec05fb', 'd...		
	6270NFM	

NaN

6 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'bc451e2cea161c49aee8489e', 'd...		
	3953RLD	

NaN

	5030LXK	
NaN		
	5534TPH	{'destination':
'94dc4a481202c4e834d9b1b1', 'd...		
	6270NFM	

NaN

7 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'6b917e5200c18cc7dc5706e2', 'd...		
	3953RLD	

NaN

	5030LXK	
NaN		
	5534TPH	{'destination':
'69527a1b66c815a1c584ceb2', 'd...		
	6270NFM	

NaN

8 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'dc1249deb34751bea752d8c6', 'd...		
	3953RLD	

NaN

	5030LXK	
NaN		
	5534TPH	{'destination':
'94a4877361dfc7ba1b87cab4', 'd...		
	6270NFM	

NaN

9 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination': '6c7bae4b49d462a11e89e841', 'd...
	3953RLD	
NaN		
	5030LXK	
NaN		
	5534TPH	{'destination':
'314caa09135c1706043b68e8', 'd...	6270NFM	
NaN		

10 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination': '5aeeca2cdb6fe31768e2482', 'd...
	3953RLD	
NaN		
	5030LXK	
NaN		
	5534TPH	{'destination':
'12416a3a52d608766c9e994d', 'd...	6270NFM	
NaN		

11 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination': '93b83fa292469c49f341ee8c', 'd...
	3953RLD	
NaN		
	5030LXK	
NaN		
	5534TPH	{'destination':
'ce6feed06bc1a9e5aff95f04', 'd...	6270NFM	
NaN		

12 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination': '26e11a1c88e779b5db8896c9', 'd...
	3953RLD	
NaN		
	5030LXK	
NaN		

	5534TPH
NaN	
	6270NFM
NaN	

13 \

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'c0ff0c4e5f60828dfc8194ab', 'd...		
	3953RLD	
NaN		
	5030LXK	
NaN		
	5534TPH	
NaN		
	6270NFM	
NaN		

14

simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	{'destination':
'ce6feed06bc1a9e5aff95f04', 'd...		
	3953RLD	
NaN		
	5030LXK	
NaN		
	5534TPH	
NaN		
	6270NFM	
NaN		

```
[53]: conjunto[["eventTime", "eventType"]].explode(["eventTime", "eventType"]).head()
```

```
[53]:
```

		eventTime \	
simulationId	truck_id		
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	660	
	3321FBL	751620	
	3321FBL	751620	
	3321FBL	872250	
	3321FBL	872250	

		eventType
simulationId	truck_id	
007f3d08-14a7-4a93-af9e-f0d9cfbcea94	3321FBL	Truck departed
	3321FBL	Truck started delivering
	3321FBL	Truck arrived
	3321FBL	Truck started delivering

3321FBL Truck ended delivering

```
[14]: # planes = planes.join(planes.route.explode().apply(pd.Series), lsuffix='_sim').  
      ↪reset_index(drop=True).drop("route", axis=1)  
      # planes
```

Exploro la columna items_sim y me doy cuenta de que los items pueden estar en diferentes localizaciones (Esta línea me ha tardado poco más de un minuto en ejecutar)

```
[85]: # planes.items_sim.explode().apply(pd.Series).locationId.apply(lambda x: x in  
      ↪planes.destination.to_numpy()).all()
```

```
[ ]:
```